

Discussion on the impact of industry 4.0 and Industry 5.0 regarding AI governance

The relationship between artificial intelligence (AI) and sustainability within Industry 4.0 is often framed in overly optimistic terms, emphasising efficiency gains, automation, and improved decision-making. However, this perspective risks underestimating both the structural limitations of AI and the broader societal implications of its deployment. As noted by Schwab (2017), technological development in the Fourth Industrial Revolution is advancing at an unprecedented pace, often outstripping the capacity of institutions to adapt.

On the one hand, AI undoubtedly offers significant potential benefits. Advanced data processing and modelling techniques enable the analysis of complex systems at a scale that would be impossible through human labour alone. In the context of sustainability, this can support climate modelling, resource optimisation, and more informed policy design. As Metcalf (2024) argues, computational systems are particularly valuable in handling large-scale, data-intensive challenges where traditional approaches fall short. From this perspective, AI can be seen as a necessary tool in addressing global issues such as climate change and environmental degradation.

However, this techno-optimistic view becomes problematic when it implicitly assumes that better data and more advanced models will translate into better decisions. Metcalf (2024) explicitly challenges this assumption, noting that while AI can assist in analysing complex problems, it cannot resolve the fundamentally social, political, and economic trade-offs that define sustainability transitions. This aligns with broader critiques in the data science literature, which emphasise that data-driven systems are not neutral but embedded within social and institutional contexts (Kitchin, 2014). In other words, AI may improve our understanding of problems, but it does not eliminate conflicts of interest or value-based disagreements.

Moreover, the risks associated with AI extend beyond technical limitations to systemic and societal concerns. A key issue is growing dependency. As technological systems become increasingly embedded in critical infrastructures and decision-making processes, the ability to function without them diminishes. Metcalf (2024) highlights a scenario in which humans retain nominal control over machines, yet become so dependent on them that disengagement is no longer a viable option. This represents a subtle but important shift: the problem is not necessarily that AI systems “take over,” but that human autonomy is gradually eroded through reliance.

In addition, concerns around autonomy and control—while sometimes framed as speculative—raise important governance questions. These concerns are increasingly reflected in policy discussions, including the European Commission’s efforts to regulate AI through a risk-based framework (European Commission, 2021). However, such regulatory initiatives often lag behind technological development, particularly in specialised domains such as official statistics, where clear guidance on issues like cloud usage and data governance remains limited.

From a critical perspective, the central challenge is therefore not simply how to use AI effectively, but how to govern it responsibly. As Schwab (2017) suggests, the transformative power of new technologies requires equally transformative approaches to governance. Without robust regulatory frameworks and institutional oversight, there is a risk that AI systems will reinforce existing inequalities and prioritise efficiency over accountability.

In conclusion, while AI has an important role to play in addressing complex global challenges, its contribution should not be overstated. A purely technological solution to inherently social problems is unlikely to succeed. A more balanced approach requires recognising both the capabilities and the limits of AI, and placing greater emphasis on governance, human oversight, and institutional responsibility.

References

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